Course: CIVE 202 - Civil Engineering Analysis II | Spring 2026

Assignment: Project 4, Risk Analysis

Client: Risk Averse, LLC

Due Date: 4/21/2026 at 5:00p.m.

Project Engineers: Group F (Macy D., Aminasahra W., Arturo G.)

Introduction

This report summarizes an analysis of the Federal Emergency Management Agency's (FEMA) National Risk Index (NRI), performed by Group F on behalf of Risk Averse, LLC. As natural disasters increase in severity and frequency, the methods used to quantify community risk become crucial for policymaking and allocation of resources.

The main goal of this study is to do a sensitivity analysis on the NRI dataset. By making our own alternative risk scoring definition and applying it to Census Tract data for Louisiana and Virginia, our team aims to identify potential biases. This analysis examines if current NRI rankings unintentionally put specific communities at a disadvantage due to the heavier weighting of certain variables over others.

Through the integration of the NRI dataset and the CDC's Social Vulnerability Index (SVI), this report provides a comparative look at how risk definitions affect disaster preparedness policy and future community resilience.

Methods

1. Data Selection and Preparation

We pulled Census Tract data for Louisiana and Virginia from the FEMA NRI (2025) and the CDC/ATSDR SVI (2022) databases and imported them to Python.

- Cleaning: We removed any Census Tracts with missing values for Expected Annual Loss (EAL) or Community Resilience (CR) to ensure the final rankings were not skewed by incomplete data.
- Reproducibility: We used the raw score values (not the percentile rankings) for EAL, SVI, and CR to perform our calculations.

2. The Calculation

The standard NRI uses a multiplicative approach (this is explained in depth in the Risk Definition Comparison). We replaced this with an additive approach to ensure social factors are not counted out by low property values.

Our calculation follows this:

1. Resilience Scaling: Divide the Expected Annual Loss (EAL) by Community Resilience (CR). This calculates the damage remaining after a community's internal resources are used.
2. Social Integration: Add the Social Vulnerability Index (SVI) score to the result.

3. Comparison Logic

After calculating the new scores, we re-ranked all Census Tracts within each state. We then compared these new rankings to the original NRI rankings to identify which tracts positions changed significantly. A large increase in rank under our formula indicates a community that is highly vulnerable but was previously overlooked due to their low property values.

Risk Definition Comparison

The NRI's current definition of risk is $Expected\ Anual\ Loss * \frac{Social\ Vulnerability}{Community\ Resillience}$. We propose changing it to $\frac{Expected\ Annual\ Loss}{Community\ Resillience} + Social\ Vulnerability$. This new risk calculation redefines the relationship between physical loss and societal factors by changing the original multiplication formula to an additive approach. This method addresses and attempts to help eliminate bias for the wealthy. Our group is updating this risk formula to ensure that areas in need are not overlooked just because they have lower property values. Instead of multiplying every factor together, we are using a method that treats social needs as a separate, added value. To start, we take the expected annual loss (EAL) from a natural disaster and divide it by the community's resilience score. This step shows how much actual damage is likely to remain after a town tries to recover on its own. Next, we add the social vulnerability score directly to that result rather than using it as a multiplier. This ensures that a community with high social needs will always show up at high risk, even if the buildings there aren't that expensive.

The implications of this change would be significant for federal natural disaster policy/resources.

- Ethical Resource Allocation: By moving away from a multiplicative model, we ensure that Social Vulnerability is no longer diluted by low property values. In the current NRI, a wealthy area with high-value real estate often ranks at a higher risk than a low-income area, simply because the amount of potential money loss is greater. Our method ethically moves risk around human need rather than just asset replacement.

- FEMA Funding: Since NRI scores are used to prioritize FEMA grants for building resilient infrastructure, our definition would likely shift the flow of federal funds toward historically underfunded Census tracts. This allows communities with lower tax bases to finally access the funding needed for preparing/recovering from natural disasters.
- Policy and Interpretation: States using our proposed index would be encouraged to make decisions and policies that prioritize vulnerable populations during the planning phase, rather than only reacting to the highest dollar-value damage sites.

Results

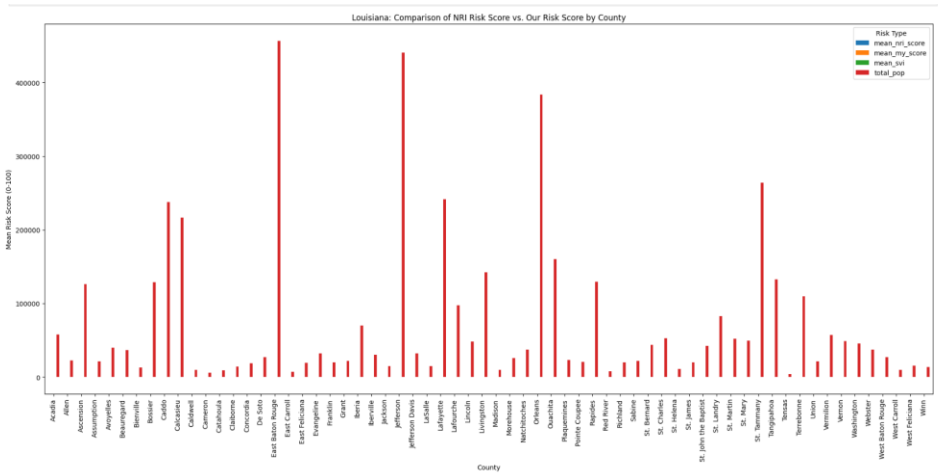

Figure 1 – Louisiana Population Distribution by County

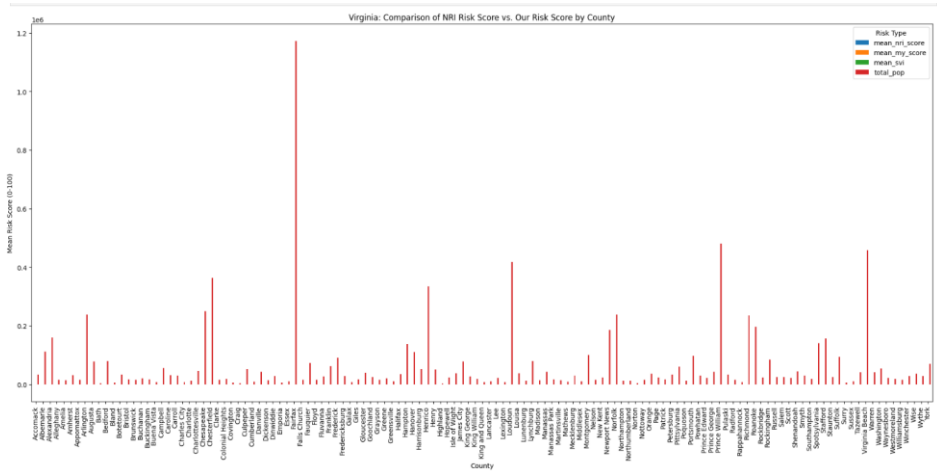

Figure 2 - Virginia Population Distribution by County

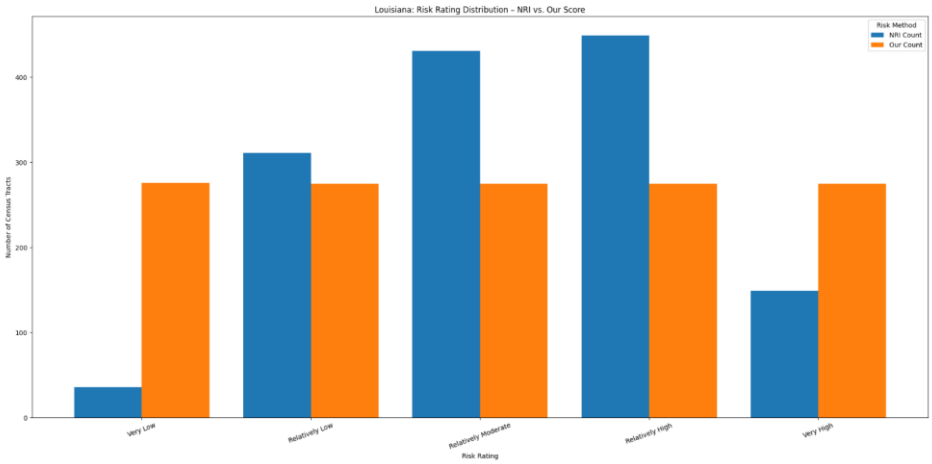

Figure 3 – Louisiana Risk Rating Distribution (NRI (Blue) vs Our Score (Orange))

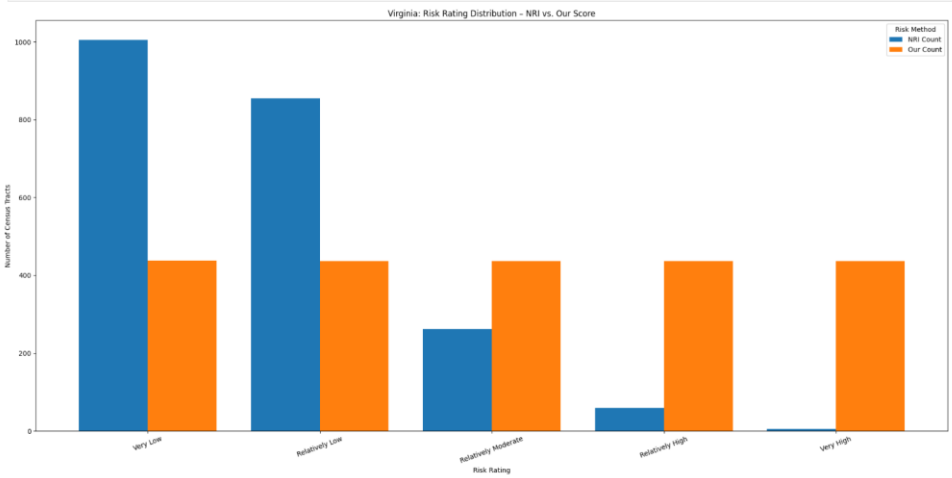

Figure 4 – Virginia Risk Rating Distribution (NRI (Blue) vs Our Score (Orange))

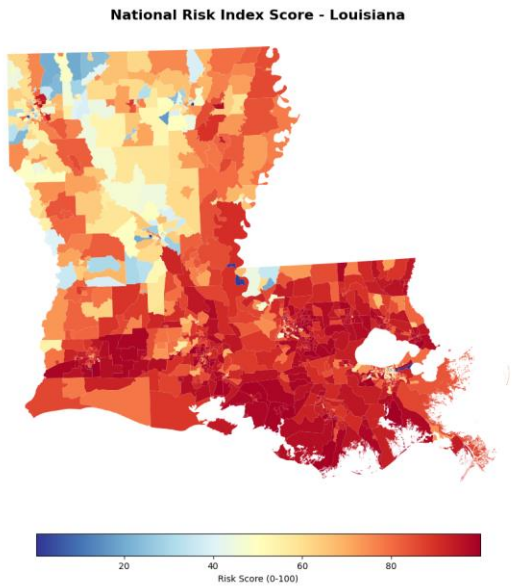

Figure 5 - Louisiana National Risk Index Map

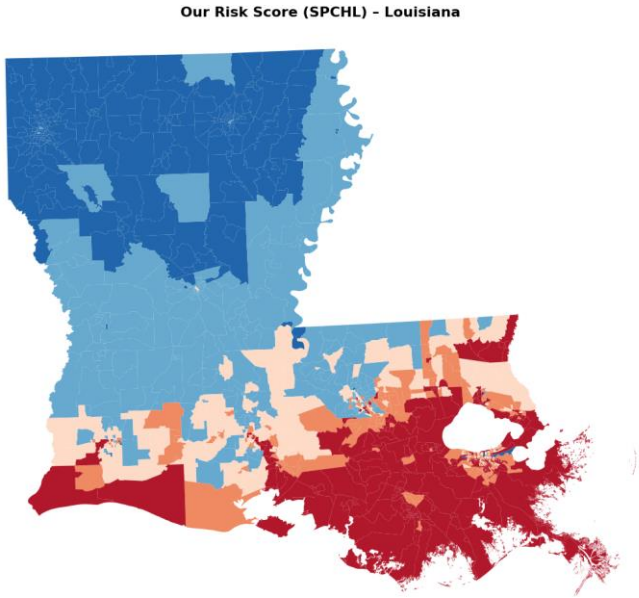

Figure 6 - Louisiana Map Based on Our Risk Formula

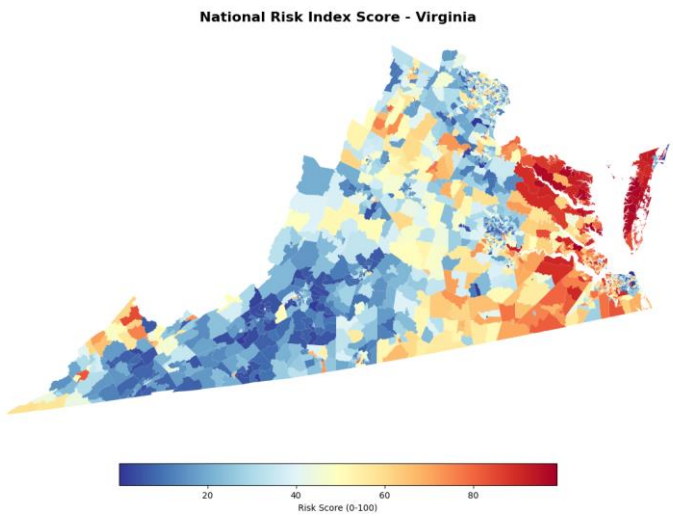

Figure 7 - Virginia National Risk Index Map

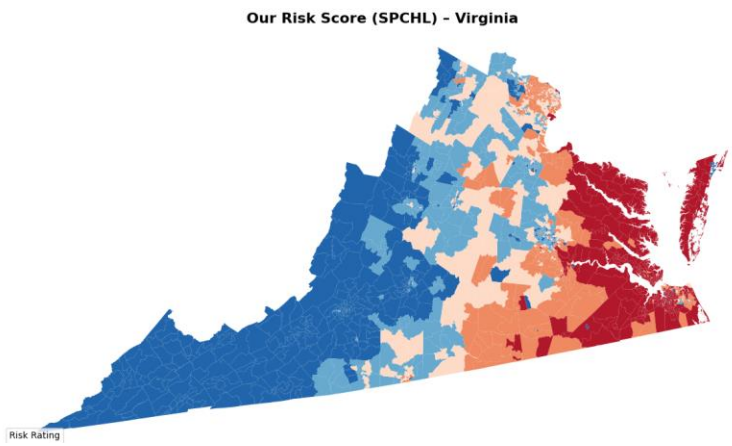

Figure 8 - Virginia Map Based on Our Risk Formula

Results Analysis

The comparison between the NRI definition and our additive formula shows a significant shift in which counties are classified as at risk. By moving away from a model using multiplication, we removed the part of the calculation that often limits low-income areas from being classified as high priority.

- County-Level Comparisons (Figures 1 and 2): These charts show the population distribution by county in both Louisiana (Figure 1) and Virginia (Figure 2). Counties in Louisiana with very high populations include East Baton Rouge, Jefferson, and Orleans. The county in Virginia with the largest population of a lot is Fairfax.
- Risk Rating Distributions (Figures 3 and 4): In Louisiana (Figure 3), the NRI distribution is skewed toward "Relatively Moderate" and "Relatively High." Our score, however, creates a uniform distribution across all categories. This indicates that the NRI may be

overconcentrating risk scores in specific high value areas, while our model shows a wider range of communities in need of support.

  - In Virginia (Figure 4), the NRI heavily skews toward "Very Low" risk. Our formula corrects this by identifying "Very High" risk zones that the standard NRI fails to account for, most likely due to lower property density in those areas.

- Louisiana Choropleth Maps (Figures 5 and 6): The NRI map (Figure 5) concentrates on risk pretty well distributed with a focus on coastal areas. Our map (Figure 6) shows that when Social Vulnerability is added as a main component, the risk gets more isolated into the coastal Southeast regions of Louisiana.
- Virginia Choropleth Maps (Figures 7 and 8): Like Louisiana, the Virginia NRI map (Figure 7) identifies moderate to high risk in coastal zones but also across other portions of Northern Virginia, the Richmond area, and parts of the Southwest. On the other hand, our alternative risk map (Figure 8) shows a large concentration of risk along the Atlantic coast and the Eastern shore, with almost all non-coastal areas, receiving lower alternative risk scores.

Conclusion

This analysis shows that changing the NRI risk definition to a different formula shifts which communities are labeled as high risk and are prioritized. While our additive approach successfully gets rid of some of the wealth bias, it creates a big difference in prioritizing coastal regions over more inland areas. This could lead to inland areas receiving less aid, however coastal areas are more commonly/severely impacted by natural disasters typically.

Impact on Communities and Funding

- Geographic Equity vs. Social Need: By focusing on risk so intensely on the coast, our formula of risks overlooking inland communities in Louisiana and Virginia. These areas may have high social needs but lack the extreme environmental hazard data of the coastline, creating geographic bias.
- FEMA Funding Allocation: If used for FEMA grants, our model would likely direct almost all federal mitigation dollars to coastal projects. While this does a good job protecting high hazard zones, it adds a barrier for vulnerable inland areas to access the same lifesaving infrastructure funding.
- Policy Recommendations: We recommend that Risk Averse, LLC advise clients that no single formula is bias-free. Our results show that while the additive approach helps solve the wealth bias of property values, it can unintentionally create geographic bias. Future policy should use multiple approaches that evaluate both coastal environmental threats and inland social vulnerability independently and find a way to combine it in order to properly allocate natural disaster resources.

References

FEMA National Risk Index. (2025). *National Risk Index: Understanding Scores and Ratings.* Federal Emergency Management Agency. Retrieved from: https://hazards.fema.gov/nri/understanding-scores-ratings

FEMA National Risk Index. (2025). *Data Resources: Census Tract Level Data (Louisiana & Virginia).* Federal Emergency Management Agency. Retrieved from: https://hazards.fema.gov/nri/data-resources

Agency for Toxic Substances and Disease Registry (ATSDR). (2022). *Social Vulnerability Index (SVI) Data and Documentation.* Centers for Disease Control and Prevention. Retrieved from: https://www.atsdr.cdc.gov/place-health/php/svi/svi-data-documentation-download.html

FEMA National Risk Index. (2025). *Technical Documentation and Metadata: Data Dictionary.* Federal Emergency Management Agency. Retrieved from: https://hazards.fema.gov/nri/data-resources

Google. (2026). Google Gemini [Large Language Model] https://gemini.google.com

Appendix: AI Teaching Assistance Disclosure

Prompt Used: "Act as a data analyst. I am analyzing the NRi and SVI for 2 states in Python. Help me understand and figure out how to build a combined risk score which is called SPCHL, I am completely lost and cannot figure it out and I tried alternative routes and kept not coming out as I want it to come out. How can this be done better, while this is being given, tell me each line of how it works and what is better from it than my code that I made." *Used on April 19th, 2026* This is where it started teaching me how to improve and why it started working and giving out a better result than what I kept getting, which was frustrating. I for one hate AI just simply like yes, it makes things easy, but I hate how immediately most will not even attempt to do anything and go straight to AI which is the worst. I feel like it truly brings humanity down as a whole with the dependency on AI. I am more for one to use as a tool for teaching in case I truly cannot understand it like, for example, Calculus II. I had a difficult time understanding the series and the MSLC was not helpful at all. So, I asked AI to teach it to me like I am brand new to Calculus and went ahead and mastered it by making quizzes and problems of it. Now for something like python, I mainly want to learn from it and not be like others where it's an immediate copy and paste. It's more like a learning tool to help when you are struggling. Now the 2 photos below are what helped me develop and walk me step by step as to what to improve on. Thank you.

```
# Louisiana: Build the combined SPCHL score with the weighted combination; 40% hurricane, 40% flood, 20% tornado
la_complete['combined_risk'] = (0.40 * la_complete['hurricane_risk'] +
                                0.40 * la_complete['flood_risk'] +
                                0.20 * la_complete['tornado_risk'])
la_complete['risk_per_capita'] = la_complete['combined_risk'] / (la_complete['POPULATION'] + 1)
la_complete['my_risk'] = la_complete['risk_per_capita'] * (1 + la_complete['RPL_THEMES'])
la_min = la_complete['my_risk'].min()
la_max = la_complete['my_risk'].max()
la_complete['my_risk_score'] = (la_complete['my_risk'] - la_min) / (la_max - la_min) * 100
print(la_complete['my_risk_score'])
```

```
)       0.047305
.       0.027965
!       0.026668
}       0.028585
}       0.019758
          ...
.371    0.002948
.372    0.001397
.373    0.002589
.374    0.000823
.375    0.000509
lame: my_risk_score, Length: 1376, dtype: float64
```

```
# Louisiana: Assign risk rating category using the same 5 Labels as the NRI
la_q20 = la_complete['my_risk_score'].quantile(0.20)
la_q40 = la_complete['my_risk_score'].quantile(0.40)
la_q60 = la_complete['my_risk_score'].quantile(0.60)
la_q80 = la_complete['my_risk_score'].quantile(0.80)

la_complete['my_risk_rating'] = 'Relatively Moderate'
la_complete.loc[la_complete['my_risk_score'] <= la_q20, 'my_risk_rating'] = 'Very Low'
la_complete.loc[(la_complete['my_risk_score'] > la_q20) & (la_complete['my_risk_score'] <= la_q40), 'my_risk_rating'] = 'Relatively Low'
la_complete.loc[(la_complete['my_risk_score'] > la_q60) & (la_complete['my_risk_score'] <= la_q80), 'my_risk_rating'] = 'Relatively High'

la_complete.loc[la_complete['my_risk_score'] > la_q80, 'my_risk_rating'] = 'Very High'
print('Louisiana: Our Risk Rating Distribution')
print(la_complete['my_risk_rating'].value_counts())
print('Louisiana: NRI Risk Rating Distribution')
print(la_complete['RISK_RATNG'].value_counts())
```

```
# Virginia: Create risk columns, somewhat same steps as Louisiana
va_complete['hurricane_risk'] = va_complete['HRCN_AFREQ'] * va_complete['HRCN_EALB']
va_complete['flood_risk']     = va_complete['CFLD_AFREQ'] * va_complete['CFLD_EALB']
va_complete['tornado_risk']   = va_complete['TRND_AFREQ'] * va_complete['TRND_EALB']
va_complete['combined_risk']   = (0.40 * va_complete['hurricane_risk'] +
                                  0.40 * va_complete['flood_risk'] +
                                  0.20 * va_complete['tornado_risk'])

va_complete['risk_per_capita'] = va_complete['combined_risk'] / (va_complete['POPULATION'] + 1)
va_complete['my_risk'] = va_complete['risk_per_capita'] * (1 + va_complete['RPL_THEMES'])
va_min = va_complete['my_risk'].min()
va_max = va_complete['my_risk'].max()
va_complete['my_risk_score'] = (va_complete['my_risk'] - va_min) / (va_max - va_min) * 100
va_q20 = va_complete['my_risk_score'].quantile(0.20)
va_q40 = va_complete['my_risk_score'].quantile(0.40)
va_q60 = va_complete['my_risk_score'].quantile(0.60)
va_q80 = va_complete['my_risk_score'].quantile(0.80)

va_complete['my_risk_rating'] = 'Relatively Moderate'
va_complete.loc[va_complete['my_risk_score'] <= va_q20, 'my_risk_rating'] = 'Very Low'
va_complete.loc[(va_complete['my_risk_score'] > va_q20) & (va_complete['my_risk_score'] <= va_q40), 'my_risk_rating'] = 'Relatively Low'
va_complete.loc[(va_complete['my_risk_score'] > va_q60) & (va_complete['my_risk_score'] <= va_q80), 'my_risk_rating'] = 'Relatively High'
va_complete.loc[va_complete['my_risk_score'] > va_q80, 'my_risk_rating'] = 'Very High'

print('Virginia: Our Risk Rating Distribution')
print(va_complete['my_risk_rating'].value_counts())
print('Virginia: NRI Risk Rating Distribution')
print(va_complete['RISK_RATNG'].value_counts())
```